



Armada Kalamunda Group

Upper Gastrointestinal Bleeding Management Guideline

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1. Purpose

This guideline documents the clinical care of patients presenting with upper gastrointestinal bleeding Upper Gastrointestinal Bleeding (**UGIB**) at Armadale Health Service Emergency Department

2. Applicability

The information in this guideline is applicable to all clinical staff working within the Emergency Department at Armadale Health Service

3. Guidelines

Bleeding from the upper GI tract is a medical emergency and is associated with significant morbidity and mortality, especially in the elderly population

Most common causes of acute UGIB are non-variceal (**NVUGIB**), which includes peptic ulcers, mucosal erosive disease, Mallory-Weiss syndrome, and malignancies

Early recognition, resuscitation, pharmacologic management, and appropriately timed endoscopic intervention are vital in the mitigation of morbidity and mortality

3.1 Initial assessment of the patient presenting with haematemesis and/or melaena

3.1.1 Immediate management of haemodynamically unstable patient

- Ensure patient in resus bay, call for help as appropriate
- If active haematemesis ensure full protective PPE for staff i.e. gown, gloves, face shield (or mask and eye protection)
- Airway: protect airway, suction (consider double suction set-up and meconium aspirator available), adjuncts/definitive airway as required
- Breathing: minimise risk of aspiration, maintain saturations >94%, give high-flow oxygen via non-rebreather mask
- Circulation:
 - obtain 2 wide bore IV lines ($\geq 18G$), ideally in large veins e.g., antecubital fossa
 - obtain urgent bloods: VBG, FBC, U&E, LFTs, Coags, Cross match, ROTEM® (additional citrate tube)
 - commence fluid resuscitation with warmed crystalloid fluids (avoid large saline load) and activate [Massive Transfusion Protocol \(Non-Obstetric\)](#)
 - Tranexamic acid is **NOT recommended** in UGIB – there is no reduction in 5-day mortality with TXA and may be associated with a small increased risk of venous thromboembolic events in these patients¹
 - consider arterial line for invasive blood pressure monitoring

3.1.2 Further assessment

- History: duration of symptoms including symptoms of collapse or dizziness
- Risk Factors for UGIB:
 - alcohol use
 - liver disease
 - previous upper GI bleed

- coagulopathy
- medications (antiplatelet or anticoagulants, steroids, NSAIDS)
- Risk factors indicating possibility of variceal bleeding:
 - history of excessive alcohol intake
 - previous variceal bleed or known history of varices
 - presence of known liver disease
 - presence of signs of portal hypertension
 - abnormal liver function test (LFT)
 - abnormal international normalised ratio (INR)
 - presence of hypersplenism with leucopenia +/- thrombocytopenia
- Medical risk factors of severity:
 - cardiac failure
 - chronic kidney disease
 - chronic lung disease
 - chronic liver disease
- Physical examination:
 - General exam including looking for features of chronic liver disease
 - Rectal exam for the presence of melaena
 - Lying and sitting blood pressure (important to detect early signs of shock >15mmHg systolic drop)

3.1.3 Investigations

- Bedside: BSL, ECG (ischaemic changes very common)
- Laboratory:
 - VBG, FBC, U&E, LFTs, Coags, G+H/Cross match/ROTEM® (as required)
 - Note initial Hb can be misleading in acute blood loss
 - Take blood, urine (and ascites) cultures in patients with liver disease
- Radiological: as indicated e.g., Erect CXR (pneumoperitoneum from perforated gastric ulcer, pneumomediastinum from oesophageal rupture, aspiration etc)

3.2 Calculation of Glasgow Blatchford Score (GBS)

ALL patients presenting with UGIB should have a [Glasgow Blatchford Score](#) calculated AND documented in their medical record

The GBS score predicts the risk of mortality and intervention at endoscopy. It is a tool that aids in decision making, but does not replace the need for ongoing clinical assessment

Refer to [Appendix 1: Calculation of Glasgow Blatchford Score](#)

- **DO NOT USE MDCALC** – this will underscore the patient
- Use information from initial assessment i.e. presenting vital signs

3.2.1 Significance of Glasgow Blatchford Score (GBS) and disposition

GBS Score	Significance
≥ 12	High risk morbidity and mortality Notify ED Consultant (contact on-call consultant overnight) Patient requires aggressive resuscitation, close monitoring, and emergency endoscopy at the earliest opportunity (<13 hours) Urgent referral to RPH Gastroenterology (MStar overnight) Transfer via RPH ED (notify RPH ED EPIC) Keep nil by mouth
> 2 and ≤ 11	Intermediate risk Refer to RPH Gastroenterology (aim for endoscopy at RPH within 24 hours, if appropriate) Generally, these patients should not be admitted at Armadale due to lack of resources available to manage acute deterioration, especially out of hours **Individual patients who ED Consultant believes is low risk (e.g. using AIMS65 score) can be discussed on a case by case basis with the endoscopist and the admitting medical consultant for consideration of admission at Armadale (see exclusion criteria) Clear fluids only, nil by mouth from midnight prior to expected scope
2	Low risk Requires admission to hospital and aim for endoscopy within 24 hours (if appropriate) Patients presenting: - Sunday PM – Friday AM (12pm cut off) can be referred to Armadale Medical team for admission (access to scope within 24 hours – see exclusion criteria) - Friday PM – Sunday AM should be referred to RPH gastro (unable to receive scope at Armadale within 24 hours) Clear fluids only, nil by mouth from midnight prior to expected scope
≤ 1	Low risk If aged <70 AND not on anticoagulants AND competent AND sensible can be discharged on oral daily PPI with outpatient endoscopy within 1 week (eReferral to AHS Gastroenterology, RPH if exclusion criteria). To represent to ED if symptoms return If does not fit above criteria or other concerns e.g. social reasons, to refer Armadale Medical team for admission

3.3 Referral pathways

Refer to: [Appendix 2: Flowchart for patients presenting with UGIB between 0800 - 2200](#)
[Appendix 3: Flowchart for patients presenting with UGIB between 2200 - 0800](#)

3.3.1 Referral process

RPH	
0800 - 2200	Refer to Appendix 2 Flowchart RPH Gastro Registrar available via RPH Switchboard, 7 days a week, for any UGIB advice/referrals
2200 - 0800	Refer to Appendix 3 Flowchart Contact Medical Specialties Registrar (MStAR) via RPH Switchboard All High-risk (GBS $\geq$ 12) /variceal bleeds or haemodynamically unstable patients should be urgently referred to MStAR overnight, and notify Armadale ED Consultant on-call and RPH ED EPIC of transfer Consider whether stable GBS <12 patients can wait for an 0800 referral to RPH Gastroenterology Registrar
Armadale	
0800-1600 Mon - Fri	Endoscopists at Armadale are happy to discuss individual cases with ED Consultant/Registrar if required (including GBS >2 that ED believe are low risk and appropriate for Armadale endoscopy) Call Endoscopy Coordinator (9391 2475) - leave contact mobile number for endoscopist to call back on if unable to take call at that time If accepted for same day scope ED will need to complete paper inpatient endoscopy referral form If patient is > GBS 2 ED Consultant to discuss potential Armadale admission directly with admitting Medical Consultant
Low risk patients (GBS $\leq$ 1)	Patient safe for discharge Do not need to be discussed directly with Gastro unless there are specific clinical concerns – eReferral for outpatient scope
Low risk patients (GBS $\leq$ 2)	Patient needs admission Refer to the AHS Medical Team, unless exclusion criteria or cut-off time for scope at Armadale within 24 hours Medical team to make referral/booking for inpatient endoscopy at Armadale

3.3.2 Exclusion criteria for endoscopy at Armadale

The following patients are unable to be scoped at Armadale

- Paediatric patients (<16 years refer to PCH)
- End stage renal failure or end stage liver disease

- Variceal bleeds (or high-risk cases likely to require an upper GI surgeon)
- Patients with liver cirrhosis and clinically significant portal hypertension (e.g. cirrhosis and platelets < 150, known varices and/or ascites)

3.4 Pharmacological Interventions in all cases of UGIB

3.4.1 Pantoprazole

Should be commenced in all cases of GI bleeding, including where the origin of bleeding is unknown. An 80mg IV stat dose is given and can be repeated BD. 80mg BD IV bolus dosing of pantoprazole is considered equivalent to continuous infusion at 8mg/hr for 72 hours (pantoprazole 80mg in 100mL sodium chloride 0.9% at 10mL/hr)²

Bolus Dose 80mg Pantoprazole IV	
Preparation	Reconstitute 2x 40mg vials, each in 10 mL Sodium Chloride 0.9% Dilute reconstituted solution to 100 mL with compatible fluid and infuse over 15 minutes
	Note: Administration of 40mg Pantoprazole reconstituted with 10mL Sodium Chloride 0.9% can be given as IV bolus over 2 minutes

3.5 Pharmacological Interventions in suspected variceal UGIB

Discuss all suspected variceal UGIB with RPH Gastroenterology ([MStar](#) overnight) – they may recommend either Terlipressin or Octreotide depending on the patient's clinical information

3.5.1 Terlipressin

Terlipressin is **contraindicated** in patients with significant ischaemic heart disease (particularly unstable angina and recent myocardial infarction), peripheral vascular disease, chronic nephritis, and pregnancy

Bolus Dose 2mg Terlipressin Acetate	
Administration	Give 2mg of IV Terlipressin acetate (equivalent to 1.7mg of Terlipressin base) undiluted and inject slowly over 3-5 minutes

Repeat Dosing Terlipressin Acetate	
Administration	Repeat 2mg bolus dose as above every 4 hours (1mg if <50kg)
Once bleeding controlled	Reduce the bolus dose to 1mg IV Terlipressin acetate (equivalent to 0.85mg of Terlipressin base) and/or frequency to every 6 hours Diagnosis should be endoscopically confirmed (ideally performed within 12 hours of admission) and continued dosing should be confirmed by the Gastroenterologist

Note:

- Extravasation may cause injection site necrosis
- Can trigger ventricular arrhythmias – caution should be exercised in patients with concomitant medications that can prolong QT interval, or cause hypokalaemia/hypomagnesaemia

- Precaution in asthma/COPD – bronchospasm (smooth muscle constriction) may worsen

3.5.2 Octreotide

Octreotide is an alternative to Terlipressin

Bolus Dose 50 microg	
Administration	Give 50 microg IV undiluted over 2 – 3 minutes
Infusion	
Preparation	Dilute 500 microg in 500mL Sodium Chloride 0.9% (1 microg/mL)
Rate	25 – 50 mL per hour (25 – 50 microg/hour)
Fluid restriction	Dilute 500 microg in 100mL Sodium Chloride 0.9% (5 microg/mL) 5 – 10 mL per hour (25 – 50 microg/hour)

Note:

- May cause hypo or hyperglycaemia – random blood glucose monitoring is required
- May cause cardiac arrhythmias, in particular bradycardia or AV block (including complete heart block) – cardiac monitor patient

3.5.3 Antimicrobial Cover

Antibiotics should be commenced in all patients with suspected variceal bleeding and be continued for a minimum of 3 days. Prophylactic antibiotics have been shown to decrease infections and improve survival in patients with cirrhosis presenting with an upper GI bleed

Antibiotics are not required for NVUGIB patients unless concurrent sepsis

Patients in ED with suspected variceal bleeding (including immediate **non-severe** or delayed **non-severe** penicillin hypersensitivity) should receive:

- Ceftriaxone 1g IV, daily

Patients with immediate **severe** or delayed **severe** penicillin hypersensitivity should receive:

- Ciprofloxacin 400 mg IV, 12-hourly

3.6 Correction of Coagulopathy

Guidance should be sought from RPH Gastroenterology and RPH Haematology Teams

In an actively bleeding patient, with an abnormal coagulation profile, a ROTEM® should be performed to assist with blood product replacement (Armada Pathology Laboratory will consult with FSH Haematology if multiple massive transfusion packs are issued under the [Massive Transfusion Protocol](#))

In the absence of active bleeding an elevated INR does not require immediate correction

For patients taking a Direct Oral Anticoagulant (DOAC) send a Coagulation profile, anti-Xa level and drug level, and ensure early consultation with RPH Haematology

For patients taking antiplatelet agents (Clopidogrel, Prasugrel, Ticagrelor), antiplatelet effects last 3-9 days and no specific action is required to reverse effects (do not give Tranexamic Acid)

Consider platelet transfusion if platelet count < 50 x10⁹/L

- Platelets are not kept on site at Armadale (require >60-90 mins to arrive from Red Cross)

3.7 Extreme Blood Loss

If the patient is in extremis and is unstable for transfer, blood loss can usually be controlled with a [Sengstaken-Blakemore-type tube](#) for 24 to 48 hours. This should ONLY be done in conjunction with intubation, with senior consultant or anaesthetic support

4. Legislative context

The following documents are required to give effect to this policy:

- Health Department of Western Australia (HDWA) [Recognising and Responding to Acute Deterioration Policy MP 0171/22](#)
- East Metropolitan Health Service (EMHS) Policy [Inter-Hospital Patient Transfer EMHS: 38](#)

5. Supporting information

The following documents support the implementation of this policy:

- [Medication Management Procedure AKG-PRO-0255-22](#)
- [Recognition and Response to Acute Deterioration Policy AKG-POL-0388-22](#)
- [Blood Management Manual AK-0626-23](#)
- [AKG Massive Transfusion Protocol \(Non-Obstetric\)](#)
- [Sengstaken-Blakemore Tube – Obtaining Haemostasis Procedure THE-PRO-0411-13](#)

6. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 1 – Action 4.01 Integrating clinical governance
- Clinicians use the safety and quality systems from the Clinical Governance Standard when:
 - a. Implementing policies and procedures for medication management
 - b. Managing risks associated with medication management
 - c. Identifying training requirements for medication management
- Standard 7 – Action 7.06 Prescribing and administering blood and blood products
- The health service organisation supports clinicians to prescribe and administer blood and blood products appropriately, in accordance with national guidelines and national criteria
- Standard 8 – Action 8.04 Recognising acute deterioration
- The health service organisation has processes for clinicians to detect acute physiological deterioration that require clinicians to:
 - a. Document individualised vital sign monitoring plans
 - b. Monitor patients as required by their individualised monitoring plan
 - c. Graphically document and track changes in agreed observations to detect acute deterioration over time, as appropriate for the patient
- Standard 8 – Action 8.06 Escalating care
- The health service organisation has protocols that specify criteria for escalating care, including:
 - a. Agreed vital sign parameters and other indicators of physiological deterioration

- b. Agreed indicators of deterioration in mental state
- c. Agreed parameters and other indicators for calling emergency assistance
- d. Patient pain or distress that is not able to be managed using available treatment
- e. Worry or concern in members of the workforce, patients, carers, and families about acute deterioration

7. References

1. Effects of a high-dose 24-h infusion of tranexamic acid on death and thromboembolic events in patients with acute gastrointestinal bleeding (HALT-IT): an international randomised, double-blind, placebo-controlled trial. The Lancet. Vol 395, issue 10241, p1927-1936, June 20, 2020. DOI: [https://doi.org/10.1016/S0140-6736\(20\)30848-5](https://doi.org/10.1016/S0140-6736(20)30848-5)
2. Hamita Sachar et al, Intermittent vs continuous proton pump inhibitor therapy for high-risk bleeding ulcers: a systematic review and meta-analysis, AMA Intern Med. 2014 Nov; 174(11): 1755–1762. DOI: <https://doi.org/10.1001/jamainternmed.2014.4056>
3. [RPH Non-Variceal Upper Gastrointestinal Bleeding \(NVUGIB\) Management Clinical Guideline](#) V3.1 (03/2024)
4. [RPH Variceal Bleeding Management Guideline](#), V3.1 (02/2023)
5. [FSFHG Upper Gastrointestinal Bleeding Guideline](#), V3 (01/2023)

8. Document owner

Enquiries relating to this guideline may be directed to:

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9. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V7.0	18/02/2025	18/02/2028	Clearer guideline for assessment, management and referral pathways between AHS and RPH

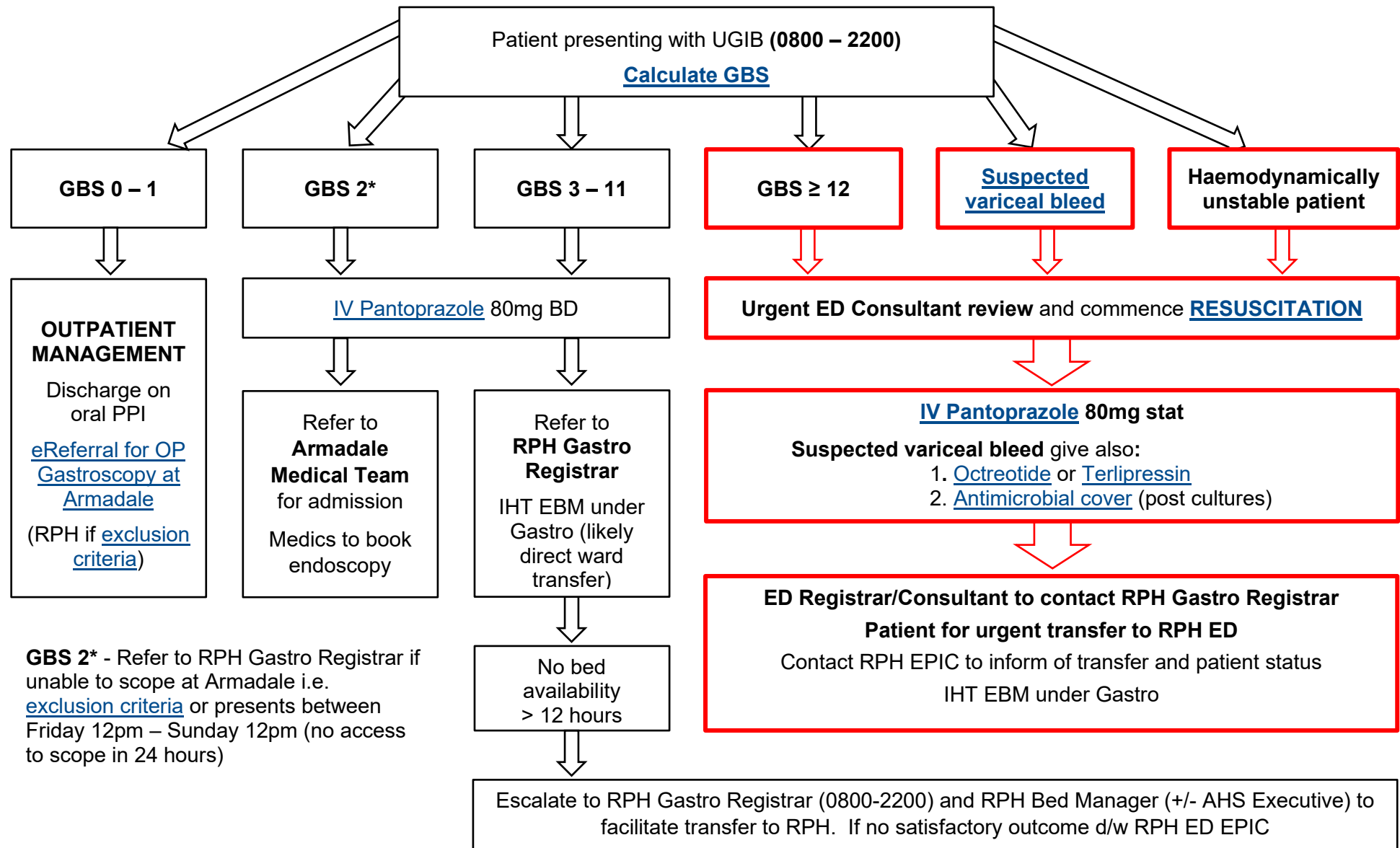
10. Authorisation

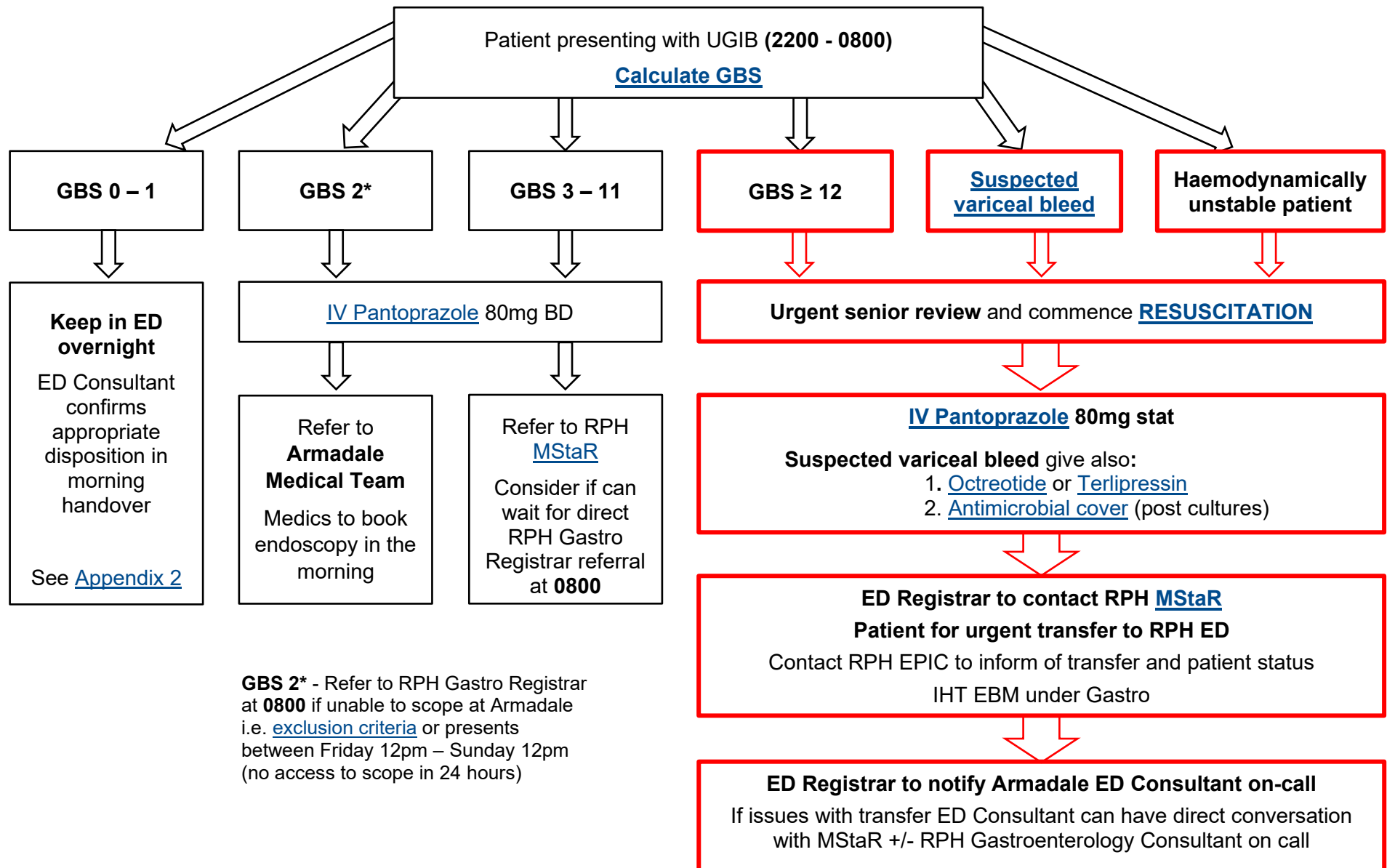
This guideline has been authorised and issued by the Director of Clinical Services.

Authorised by	Dr Tania Strickland – Director of Clinical Services
Authorisation date	18/02/2025
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Appendix 1: Calculation of Glasgow Blatchford Score

Risk marker		Score
Blood urea mmol/L	< 6.5	0
	6.5 - 7.9	2
	8 – 9.9	3
	10 – 24.9	4
	> 25	6
Haemoglobin g/L Men	≥ 130	0
	120 - 129	1
	100 – 119	3
	< 100	6
Haemoglobin g/L Women	≥ 120	0
	100 – 119	1
	≤ 99	6
Systolic Blood Pressure mmHg At presentation	≥ 110	0
	100 - 109	1
	90 – 99	2
	< 90	3
Pulse At presentation	≤ 100	0
	> 100	1
Melaena At presentation	No	0
	Yes	1
Syncope At presentation	No	0
	Yes	2
Hepatic disease	No	0
	Yes	2
History of cardiac failure	No	0
	Yes	2
Maximum score = 23	Total	

Appendix 2: Flowchart for patients presenting with UGIB between 0800 - 2200

Appendix 3: Flowchart for patients presenting with UGIB between 2200 - 0800

Appendix 4: Guidance for completing outpatient endoscopy eReferral for Armadale

1. Outpatient eReferral for endoscopy is only for **patients with GBS 0 or 1 that you are discharging home**
2. Confirm outpatient eReferral is being completed for correct patient
3. Ensure appropriate contact number and ED doctor name inserted under referrer details
4. Referral Site and Unit as per example below
5. Set Urgency as **"Not Urgent"** – this means you do not need to insert a responder name or pre-discuss with endoscopist (ED UGIB endoscopy referrals will still be triaged as urgent)
6. Check that patient does not fall under exclusion criteria for endoscopy at Armadale
7. Under "Name of Gastro team member contacted" insert **"ED Referral for UGIB"**, this will ensure prioritisation on waitlist and scope in < 1 week
8. Complete Questionnaire with as much detail as available
9. Copy your ED medical notes under Additional Details tab and also submit to DMR

Referrer Details - Outpatient (Draft)		Referral To	
AKHS - Emergency Medicine Contact No: Specialist: Source of Referral: Emergency Department Reason for Referral: Assessment Validity Period: 3 Months		Site: Armadale Health Service Unit: Gastroenterology [GAS] Responder: Urgency: Not Urgent  referral reviewed within 5 business days	
Hints to Unit/Service For Direct Access Endoscopy Only for Armadale & Kalamunda. Patient will be allocated to Armadale or Kalamunda during clinical triage. Please refer to the Unit Gastroenterology only and we will allocate. Clinician Assist WA » Endoscopy Requests Referrals without the essential information will be returned For REPEAT SCOPES please select the service for Repeat Scopes: Armadale Exclusions:		Details Questionnaire* ED Referral for UGIB Procedure Requested: ☐ Colonoscopy ☐ Flexible Sigmoidoscopy ☐ Gastroscopy ☐ Gastroscopy & Colonoscopy ☐ Gastroscopy + PEG ☐ Other ☐ Abdominal pain ☐ Abnormal imaging Indications:	